3. Handling Missing Data

	Age	Ht	Wt	Exp	Salary	
~~100~~	26	5.5	58		19K	×

0 df.isnull().sum()

For handling null values :-

- -> Drop the row
- -> Impute / Fill that null value.

The choice depends on some criterias :-

-> how much data is missing
-> Type of the feature.
-> Distribution of the data.
-> Reason behind missingness (

CCA (Complete Case Analysis)

Removing every row containing atleast one missing value.

missing value.

when

✓ -> Missing values are very small in number (less 5%)

✓ -> Dataset should be large enough that losing few rows will be acceptable.

✓ -> Missing values at Random.

dropna

```
df = df.dropna()
```

How to impute your null values?

1) Mean imputation
2) Median imputation
3) Mode imputtion.
4) Arbitary values
5) Backward fill
6) Forward fill

```
df['age'] fillna(df['age'].mean())
```

c) Forward .

Mean Imputation

```
df['Age'] = df['Age'].fillna(df['age'].mean(), inplace=True)
```

Age
10
20
11
12
15
-
-
16
19

-> Column is numerical
-> Data is roughly normally distributed.
-> No extreme outliers.

Btech 1st year class

Students - 19 - 22 <- 20.5

Teacher - 55 35

Two students

35 35

Median Imputation

we will just take median imputation after sorting.

when

-> Column is numerical
-> Distribution is skewed
-> Dataset contains outliers.

Mode

→ Replacing Null value with most frequently occuring values.

Arbitary Value Imputation

Age	
20	
21	
22	
—	26
23	25
23	
24	

bfill()

ffill()

```
df['Age'] = df['Age'].fillna(26, inplace=True)
```